

Syllabus

Spring semester 2026

Course Title	Probability and Statistics		Course Code	MC3108	Credit	3
					Hours	4
Teacher	Name	Class no.	Classroom	Office hours*	Office (Tel) E-mail	
	박윤주 Park Yunju	KC 1	3406	Mon 6,7 Wed 5,6	3504 (051-606-2387) parkyunju@ksa.hs.kr	
	김종민 Kim Jongmin	EC 1	3404	Mon 6, Tue 6 Thu 6	3507 (051-606-2149) kimjongmin@ksa.hs.kr	

*Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Descriptions

- ♦ To Introduce students to applying probability theory to gain insight into real-world statistical problems and situations.
- ♦ To introduce students to understand the method of data collection, its subsequent description, and its analysis.

2. Core Target Competencies

Subject Core Competencies			
Problem solving ability	Reasoning	Communication	Attitude
♦ Problem solving ability : The ability to use mathematical knowledge to explore solution strategies and find solutions ♦ Reasoning : The ability to guess mathematical facts, analyze and justify them logically, and reflect on the process ♦ Communication : The ability to express mathematical knowledge or ideas, problem-solving processes in words, writing, or symbols, and to understand other people's ideas ♦ Attitude : The ability to recognize and utilize the value of mathematics based on a self-directed learning attitude			

3. Textbook:

Introduction to Probability and Statistics for Engineers and Scientists, 6th ed., Sheldon M. Ross, Elsevier Academic Press, 2021.

4. Evaluation

A. Evaluation Criteria

Midterm (CH. 2, 3, 4, 5)	Final (CH. 6, 7, 8, 9)	Homework & Presentation	Attendance & Attitude	Total
35 %	45 %	10 %	10 %	100 %
140 pts	180 pts	40 pts	40 pts	400 pts

B. Evaluation Method

- ♦ The final grade is determined by absolute evaluation.
- ♦ Homework & Presentation
 - For each chapter, the homework problems will be assigned.
 - You will make a presentation for solving homework problems.
 - Homework List

CH. 3 : 11, 17, 21, 22, 23, 27, 28, 30, 33, 35, 40, 41, 43, 45
CH. 4 : 4, 10, 12, 14, 16, 25, 29, 30, 35, 37, 44, 46, 49, 50, 56, 57
CH. 5 : 9, 16, 20, 23, 26, 29, 33, 35, 36, 37, 45, 47
CH. 6 : 2, 4, 6, 13, 15, 17, 20, 25
CH. 7 : 1, 2, 4, 5, 7, 8, 11, 13, 14, 15, 18, 32, 36, 40, 42, 43, 45, 57, 58
CH. 8 : 2, 6, 7, 9, 12, 15, 20, 26, 27, 29, 30, 32, 33, 38, 40, 44, 50
CH. 9 : 4, 8, 9, 10, 12, 13, 16, 26, 29

- ♦ Attendance
 - Absence –3 pts (except medical or official cases)
 - Late –1 pt

5. Criteria for Awarding Credits

- A. Attendance for at least three-quarters of the total class hours is required to earn credit.
- B. In case of an examination missed due to official leave or medical leave, the score will be calculated by multiplying [the score corresponding to the total percentile score earned by the student on other examinations in the same subject] by [100% for official leave, or 80% for medical leave]. However, if no reference score is available, a make-up examination will be conducted

6. Weekly plans

Week (period)	Topics and Contents	Note
Week 1 2.23.~3.1.	1. Introduction to Statistics 1.1. Introduction 1.2. Data Collection and Descriptive Statistics 1.3. Inferential Statistics and Probability Models 1.4. Populations and Samples	
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 2 3.2.~3.8.	2. Descriptive Statistics 2.1. Introduction 2.2. Describing Data Sets	3.2. Substitute Holiday

	2.3. Summarizing Data Sets 2.4. Chebyshev's Inequality 2.5. Normal Data Sets 2.6. Paired Data Sets and the Sample Correlation Coefficient	
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 3 3.9.~3.15.	3. Elements of Probability 3.1. Introduction 3.2. Sample Space and Events 3.3. Venn Diagrams and the Algebra of Event 3.4. Axioms of Probability 3.5. Sample Spaces Having Equally Likely Outcomes 3.6. Conditional Probability 3.7. Bayes' Formula 3.8. Independent Events	
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 4 3.16.~3.22.	4. Random Variables and Expectation 4.1. Random Variables 4.2. Types of Random Variables 4.3. Jointly Distributed Random Variables 4.4. Expectation 4.5. Properties of the Expected Value	
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 5 3.23.~3.29.	4.6. Variance 4.7. Covariance and Variance of Sums of Random Variables 4.8. Moment Generating Functions 4.9. Chebyshev's Inequality and the Weak Law of Large Numbers	
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 6 3.30.~4.5.	5. Special Random Variables 5.1. The Bernoulli and Binomial Random Variables 5.2. The Poisson Random Variables 5.3. The Hypergeometric Random Variables 5.4. The Uniform Random Variables	5.1.1 & 5.2.1. Optional Section
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 7 4.6.~4.12.	5.5. Normal Random Variables 5.6. Exponential Random Variables 5.7. The Gamma Distribution 5.8. Distributions Arising from the Normal	5.6.1 & 5.6.2. Optional Section
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	6. Distributions of Sampling Statistics 6.1 Introduction 6.2 The Sample Mean 6.3 The Central Limit Theorem 6.4 The Sample Variance 6.5 Sampling Distributions from a Normal Population	6.6 Optional Section
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 10 4.27.~5.3.	7. Parameter Estimation 7.1. Introduction 7.2. Maximum Likelihood Estimators 7.3. Interval Estimates	7.2.1 Optional Section
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 11 5.4.~5.10.	7.4. Estimating the Difference in Means of Two Normal Populations 7.5. Approximate Confidence Interval for the Mean of a Bernoulli Random Variable 7.6 Confidence Interval of the Mean of the Exponential Distribution	5.5. Children's Day
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 12 5.11.~5.17.	8. Hypothesis Testing 8.1. Introduction 8.2. Significance Levels 8.3. Tests Concerning the Mean of a Normal Population	
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 13 5.18.~5.24.	8.4. Testing the Equality of Means of Two Normal Populations 8.5. Hypothesis Tests Concerning the Variance of a Normal Population 8.6 Hypothesis tests in Bernoulli populations	
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 14 5.25.~5.31.	9. Regression 9.1. Introduction 9.2. Least Squares Estimators of the Regression Parameters 9.3. Distribution of the Estimators	5.25. Substitute Holiday
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 15 6.1.~6.7.	9.4. Statistical Inferences about the Regression Parameters 9.5. The Coefficient of Determination and the Sample Correlation Coefficient	6.3. Local Election Day
Core Competencies	Problem solving ability, Reasoning, Communication, Attitude	
Week 16 6.8.~6.14.	Make-up class week	
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	

※ The above plans may change according to the academic schedule.